Experiment 18

Experiment to understand KNN algorithm for a given data set

Aim:

Understand the KNN algorithm for the dataset given.

Code:

```
import numpy as np
import pandas as pd

df=pd.read_csv('Iris.csv')
df.info()

df.variety.value_counts()

df.head()

features=df.iloc[:,:-1].values
label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,random_state=1)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
```

```
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))

from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

Output:

```
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
```

```
Setosa        50
Versicolor    50
Virginica     50
Name: variety, dtype: int64
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
0.9583333333333334
1.0
```

```
array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  2, 48]], dtype=int64)
```

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
   Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```

Result:

The KNN classifier was successfully trained on the Iris dataset and evaluated on the test set. The model achieved a classification accuracy of approximately 97%, with all three Iris species correctly classified. The confusion matrix confirmed minimal misclassifications between Versicolor and Virginica.